BLOWUP SEQUENCES IN CLOSED POINTS CORRESPOND UNDER THE ETALE NEIGHBOURHOOD EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. The strategy section setup (elementary etale neighbourhoods) is used without further citation; representability of blowups is a standing property.

Allowed standard prerequisites (editorial).

- Representable morphisms from decent spaces. Decent Spaces, lemma representable-named-properties (cross-chapter, named). Ensures intermediate spaces Z_j are decent algebraic spaces so blowing up is defined.
- Fibres under the etale neighbourhood correspondence. lemma-equivalence-fibre (carded L26827). Identifies the closed point z_1 of the fibre over x_1 with a closed point z_{1,1} of (Z_{1,1})_{u_1}.

Source reference. Original reference 0ABT: focused-seeds/geometry/S9-spres-equivalence-sequence-blowups/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0BHA: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 415–415. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BHD: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 463–463. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X, x_i, U_i \to X, u_i$ be as in (0BHA) and assume $f : Y \to X$ corresponds to $g_i : Y_i \to U_i$ under F. Then there exists a factorization*

$$Y = Z_m \to Z_{m-1} \to \ldots \to Z_1 \to Z_0 = X$$

of f where $Z_{j+1} \to Z_j$ is the blowing up of Z_j at a closed point z_j lying over $\{x_1, \ldots, x_n\}$ if and only if for each i there exists a factorization

$$Y_i = Z_{i,m_i} \to Z_{i,m_i-1} \to \ldots \to Z_{i,1} \to Z_{i,0} = U_i$$

of g_i where $Z_{i,j+1} \to Z_{i,j}$ is the blowing up of $Z_{i,j}$ at a closed point $z_{i,j}$ lying over u_i.

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. A blowing up is a representable morphism. Hence in either case we inductively see that $Z_j \to X$ or $Z_{i,j} \to U_i$ is representable. Whence each Z_j or $Z_{i,j}$ is a decent algebraic space by Decent Spaces, Lemma 0ABT. This shows that the assertions make sense (since blowing up is only defined for decent spaces). To prove the equivalence, let's start with a sequence of blowups $Z_m \to Z_{m-1} \to \ldots \to Z_1 \to Z_0 = X$. The first morphism $Z_1 \to X$ is given by blowing up one of the x_i, say x_1. Applying F to $Z_1 \to X$ we find a blowup $Z_{1,1} \to U_1$ at u_1 is the blowing up at u_1 and otherwise $Z_{i,0} = U_i$ for $i > 1$. In the next step, we either blow up one of the x_i, $i \geq 2$ on Z_1 or we pick a closed point z_1 of the fibre of $Z_1 \to X$ over x_1. In the first case it is clear what to do and in the second case we use that $(Z_1)_{x_1} \cong (Z_{1,1})_{u_1}$ (Lemma 0BHD) to get a closed point $z_{1,1} \in Z_{1,1}$ corresponding to z_1. Then we set $Z_{1,2} \to Z_{1,1}$ equal to the blowing up in $z_{1,1}$. Continuing in this manner we construct the factorizations of each g_i.

Conversely, given sequences of blowups $Z_{i,m_i} \to Z_{i,m_i-1} \to \ldots \to Z_{i,1} \to Z_{i,0} = U_i$ we construct the sequence of blowing ups of X in exactly the same manner. $\square$